

Lab 1

Research

The Clavilux invented by Thomas Wilfred in the early 20th century is a performable instrument that generates lumia, which is understood as a visual light art at the time. The light projected onto the screen using a range of instruments including glass discs, lenses, motorized reflectors to generate shifting shapes and colours, hence resonating with the latin form of Clavilux, which means “light played by key”(Wiki)

- What inputs are available to the user? What responses does the work give?
While the original model of Clavilux doesn't allow interaction between viewers and the installation, as viewers may only watch pre-determined patterns of lumia, later versions of the installation allows interaction between user and the device.
- Who is present, and how does the piece color the relationships between them?
Only the device and the viewer(generally individuals or few people depending on the scale of the device), will be present at the scene. The relationship between installation and viewer depends on the possibility of interaction in between, since some models may not be interacted with.
- What is the piece famous for? What are its strengths and its weaknesses?
Clavilux, an instrumental way of displaying lumia, and its inventor Thomas Wilfred, demonstrate an avant garde style art form that directly influences population's understanding of what art can be and how they may be carried out.

Planning

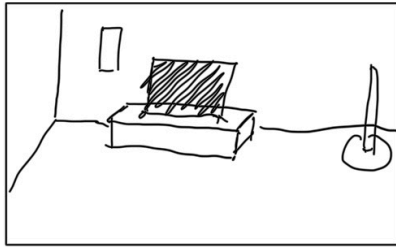
In these performances, light was the medium of expression. Therefore, it led us to consider how light can communicate intention and affect emotions. Our device explores this idea through an interactive process that reads the user's facial expressions and physical actions to generate different themes of shapes and colours with the aim of influencing their mood. The device also changes colour based on weather and time, providing users with clear visual interpretation about their surroundings. In our design the Light serves two main purposes; it creates a congenial atmosphere to influence emotion and acts as an effective medium of communication.

- Setting: The interaction is assumed to take place inside an individual's living space. We would like to interpret the setting as any where of an individual's living space that is meant to be relaxing and mindfulness.

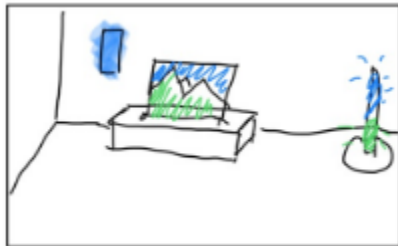
- Players: The main players in this scenario are the occupant and the lighting device. Although friends and family may occasionally visit, the general interaction process remains the same. When multiple faces are recognised the device implement changes based on an algorithm that identifies different scenarios (e.g. party, family dining and late night chat)
- Activity: The device shall be able to voluntarily collect, analyse and react based on various inputs including individual's facial expressions, body posture, sound, season, and other relevant memories to generate corresponding "lumia" at the aim of providing positive mindfulness for the recipients. The device shall accomplish its goal through altering illumination, pace, and color and with clear visual interpretation about their surroundings.
- Goals: The device generates a congenial atmosphere providing the occupant with a sense of belonging and comfort. The device also acts as an effective medium of communication generating to provide consumers with clear visual interpretation about their surroundings.

Story board

#1 Acting to environmental input



Surrounding lights are off when there is no environmental input.

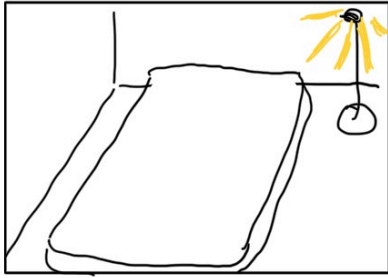


Surrounding lights may react to environmental input such as the color theme of the television content.

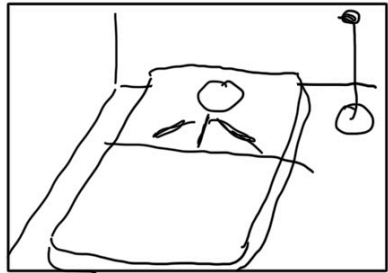


Illumination device may also react based on the content of the television, such as music, podcast, or wall paper, and choose to act accordingly.

#2 Acting to body gesture



Illumination devices may be simply turned on or off at any time with a physical switch.

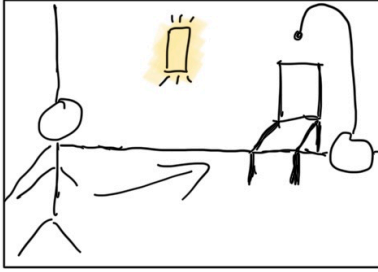


Illumination devices may be automatically reacting to input received from body gesture recognition. For example, when noticing individuals trying to sleep, the device may automatically power off.



Illumination devices may react extensively in regard to context. For example, during the middle of the night, an individual tried to get up. The device may provide optimal illumination for the purpose (going to restroom or etc)

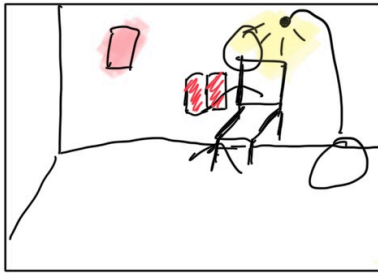
#3 Acting to Behaviour



Illumination devices may serve multiple purposes and act accordingly as it recognizes participant's purpose and behaviour. It may be used simply as a source of illumination when an individual is trying to navigate around the space.

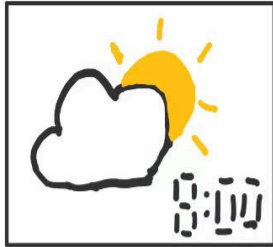


If an individual decided to proceed with another purpose, in which may benefit from having the illumination drive act differently. Like reading in this case, such behaviour may be processed through the system and sent to the illumination devices.

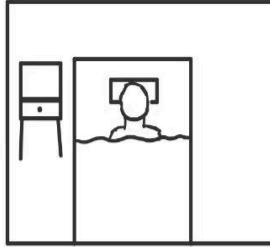


Then illumination devices may act correspondingly and mutually through adjustments of color and brightness, to better fulfill the need of the scenario.

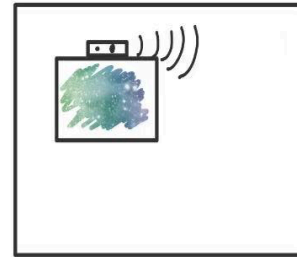
Acting to weather



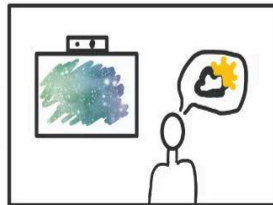
It's a cloudy day in the morning



The user wakes up in the morning wondering how is the weather



The device detects that the user is awake and display varied shape and colour based on the weather and time



The users see the lighting and knows that it's slightly cloudy